Classification of Brain Cancer Types Using Densenet-201, Resnet-50 and VGG-19 Models

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Abstract— Brain cancer is a type of serious disease that affects the vital functions of the body and is a major challenge in the medical world. This disease includes several types, such as Brain Glioma, Brain Meningioma, and Brain Tumor, which require early and accurate diagnosis to improve the patient's quality of life. This study aims to classify the three types of brain cancer using three deep learning-based convolutional neural network (CNN) architectures, namely DenseNet201, ResNet50, and VGG19. The dataset used is a T1-weighted contrast-enhanced MRI image downloaded from Kaggle, with preprocessing in the form of resizing, normalizing, and augmenting data to improve model performance. The DenseNet201 model shows the best performance with an accuracy of 99.50% and a validation accuracy of 98%, which reflects the model's ability to learn patterns efficiently without signs of overfitting. In contrast, the ResNet50 model has experienced significant overfitting. The accuracy of the training reached 37.25%, the validation accuracy was only in the range of 35%-40%, which indicates that this model is not able to capture relevant patterns in the validation data. Then the VGG19 model gives quite good results compared to ResNet50, with a validation accuracy close to 91.25% and a consistent loss trend without overfitting. The study concluded that DenseNet201 is the most effective model for the task of classifying three types of brain cancer using MRI datasets. The high performance demonstrated by DenseNet201 makes it a potential candidate for implementation in medical systems for early diagnosis.

***Keywords**—Brain Cancer, Deep Learning, DenseNet201, Medical Imaging, MRI Classification, ResNet50, Transfer Learning, VGG19.*

I. INTRODUCTION (*HEADING 1*)

Brain cancer, one of the most common and serious types of cancer, is a global health challenge due to its impact on the nervous system that regulates the body's vital functions. This cancer accounts for about 3% of all cancer cases in the world with significant geographical variation, where the incidence rate is higher in developed regions such as Europe and North America compared to developing regions such as Africa and Asia. Factors such as advances in diagnostic tools, genetic predispositions, and environmental influences contribute to these differences. The increasing trend of brain cancer cases globally shows the importance of understanding the epidemiology and its risk factors to support prevention and treatment measures[1].

Glioma, a brain tumor that originates from glial cells, is one type of brain cancer that includes glioblastoma (GBM), the most aggressive form. Gliomas often involve complex interactions between tumor cells and the microenvironment, such as the cytokine IL-33 that can recruit pro-tumorigenic macrophages and accelerate tumor growth. In general, brain tumors include a variety of abnormal growths that can be benign or malignant, with impacts such as increased intracranial pressure or damage to brain tissue. Brain tumors are difficult to treat due to the blood-brain barrier, tumor cell resistance, and the heterogeneity of the tumor's microenvironment, so recent research focuses on approaches that consider the interaction of tumors with the body's immune system. Brain Meningioma Meningioma is a type of brain tumor that originates from the meninges, the protective layer that lines the brain and spinal cord. Meningiomas are usually benign, but they can become malignant in certain cases. These tumors grow slowly and often show no symptoms in the early stages. However, as it grows, meningioma can compress brain tissue or cranial nerves, causing a variety of symptoms such as headaches, vision impairment, or weakness in the body. Treatment of meningiomas depends on the size, location, and nature of the tumor, with the main options including surgery, radiation therapy, or observation for asymptomatic cases. These tumors tend to occur more often in women and have a relationship with radiation exposure as well as certain genetic mutations such as those in the NF2 gene.

II. RELATED WORKS

The research by Tandel et al. entitled *"A Review on a Deep Learning Perspective in Brain Cancer Classification"* uses methods such as machine learning (ML) and deep learning (DL) for automatic tissue characterization and segmentation of brain tumors. The dataset includes a variety of brain images produced by imaging techniques such as MRI, CT, and PET. The study identified gaps in existing methods, such as reader-to-reader variability, reliance on pathological expertise, and data limitations, to develop more accurate and efficient computer-based tools for non-invasive brain cancer diagnosis.[1]

This study was conducted by Boeck et al. entitled *"IL-33 Drives Glioblastoma Progression by Regulating the Tumor Microenvironment"* using methods such as single-cell RNA sequencing (scRNA-seq), phosphoproteomics, in vitro functional tests, and xenograft mouse models. The dataset of this study comes from sources such as GEO (Gene Expression Omnibus) and EGA (European Genome-phenome Archive). This study fills in the gaps of previous research by exploring in depth the role of IL-33 in regulating the dynamics of microglia and macrophages in glioblastomas, which were previously poorly understood. [2]

The research by Nagaraj et al. entitled "Programmed Multi-Classification of Brain Tumor Images Using Deep Neural Network" uses a Convolutional Neural Network (CNN)-based deep learning method for multi-class classification of brain tumors. The dataset used includes public MRI images, with classification based on tumor type (Meningioma, Glioma, and Pituitary) and their grade (Grade II, III, IV). The dataset goes through a pre-processing process, including resizing to a size of 128x128x3 pixels to speed up the training process. The CNN model used consists of 16 layers with the use of ReLU activation, max pooling, and dropout functions to prevent overfitting. The study showed the highest accuracy of 96.13% for tumor type classification and 98.7% for tumor grade classification, proving the effectiveness of CNN in MRI image-based brain cancer diagnosis.[3]

The research by Vijayakumar et al. entitled *"Classification of Brain Cancer Type Using Machine Learning"* uses the Capsule Neural Network (CapsNet) to classify brain cancer types based on MRI data. This method leverages the TCGA-GBM dataset from The Cancer Imaging Archive (TCIA) and shows that CapsNet can overcome the afine rotation and transformation problems often faced by CNNs. By training using less data, CapsNet achieves higher accuracy compared to other models such as CNN, NN, and ResNet. The gaps in previous research that were addressed were the need for large datasets and the loss of input positions due to pooling in CNNs. According to CNN, the table shows the percentage accuracy of training and testing compared to other methods. However, the exact figure is like 78% or 85%. The study opens up opportunities for more efficient and accurate classification of brain cancers[4]

Research by Sadad et al. in their research entitled "Brain Tumor Detection and Multi-Classification Using Advanced Deep Learning Techniques" focuses on the detection and classification of brain tumors using a deep learning approach. The main methods used are DenseNet201 (93.1%), ResNet50 (92.9%), InceptionV3 (92.8%), and MobileNetV2 (91.8%). with additional data augmentation and transfer learning to improve model performance. The dataset used is the Figshare dataset, which includes 3,064 MRI images from 233 patients, with three main tumor types (glioma, pituitary, and meningioma) as well as three different views (sagittal, axial, coronal).[5]

Research conducted by Muhammad et al. on the multigrade classification of brain tumors using preprocessing techniques, data augmentation, transfer learning, and CNN architecture with 87% accuracy on public datasets (BraTS, TCIA, etc.), Shirazi et al. (2020) developed DeepSurvNet based on GoogleNet architecture to predict the survival rate of brain cancer patients with 99% precision using TCGA datasets and private datasets, and Sidong et al. (2020) who used a combination of GAN and ResNet50 to predict IDH mutations with 85.3% accuracy in TCGA datasets and private datasets, but these studies showed gaps such as the lack of integration of histopathological imaging in segmentation and the need to improve accuracy with additional features such as patient age.[6]

The study was conducted by Sowrirajan et al. who developed a VGG16-NADE hybrid model for brain tumor classification with a deep learning-based MRI data processing method, resulting in an accuracy of 96.01% using the T1-weighted contrast-enhanced MRI dataset, overcoming the limitations of previous models such as CNN-Genetic Algorithm or CNN-KELM in terms of sensitivity and F1 score, but there are still opportunities to improve accuracy and efficiency with more complex architectures such as VGG19.[7]

Research by Aulia and Rahmat et al. developed a brain tumor identification method based on the VGG-16 architecture and CLAHE contrast enhancement technique, with results of 90.37% accuracy, 90.22% precision, and 87.61% recall on brain MRI datasets from Kaggle, although it still faces challenges in reducing false positive predictions and improving performance in certain classes such as oligodendroglioma[8]

The study by Alshammari et al. in the year of conducting a study entitled "Construction of VGG16 Convolution Neural Network (VGG16_CNN) Classifier with NestNet-Based Segmentation Paradigm for Brain Metastasis Classification" using deep learning methods with NestNet VGG16_CNN and segmentation, achieved 93.74% accuracy, 92% precision, 92.1% recall, and 67.08% F1-score, using MRI datasets covering 3064 images, including meningioma, glioma, and pituitary tumors. With a research gap in the form of improving the diagnosis of cerebellar metastases and overcoming class imbalances in datasets through weighted softmax[9]

The research study of Tian C et al. and colleagues in 2000 developed a brain tumor classification model based on Enhanced VGG19 CNN by replacing the Softmax layer with SVM and using transfer learning, resulting in 95% accuracy through a tenfold cross-validation method to improve generalization, demonstrating superior performance in accurate and rapid diagnosis for CNS-pDLBCL and HGG.[10]

Research study of Srinivas et al. In 2010, using a deep learning-based transfer learning approach with VGG-16, ResNet-50, and Inception-v3 pre-trained models for brain tumor classification using MRI images, achieved the highest accuracy of 96% on VGG-16 models with augmentation data to increase the size of small datasets, emphasizing the efficiency of brain tumor diagnosis through automated image analysis.[11]

The research was conducted by Raza et al. and the team in 2010 developed a hybrid model of deep learning-based DeepTumorNet with GoogLeNet modification that added 15 new layers for the classification of gliomas, meningiomas, and pituitary tumors on MRI datasets, achieving 99.67% accuracy, 99.6% precision, 100% recall, and 99.66% F1-score, with an emphasis on the efficiency of automatic detection of brain tumors.[12]

The research conducted by M. Alhassan et al. developed a brain tumor segmentation and classification method based on the Bat algorithm with Fuzzy C-Ordered Means (BAFCOM) for segmentation and Enhanced Capsule Networks (ECN) for classification, achieving an accuracy of 95.81%, precision of 94.83%, recall of 94.34%, and F1-score of 94.59% on an MRI dataset of 2065 images, with a focus on improving the automatic diagnosis of brain tumors through a deep learning approach.[13]

The Mercaldo et al. research study used the YOLOv8 object detection model to detect and localize brain cancer from MRI images, achieving 94.3% precision, 93.2% recall, and 94.5% mAP@0.5 on a dataset of 300 MRI images, with an emphasis on speed and accuracy for faster and more effective diagnosis of brain cancer.[14]

The research study of Florimbi et al. developed a Hyperspectral Imaging (HSI)-based brain cancer detection system with a multi-GPU platform for intraoperative surgery, achieving a classification accuracy of 95% and a processing time of less than three seconds for the largest images in the database, ensuring a real-time response with high accuracy in distinguishing tumor, normal, and hypervascular tissues.[15]

The study conducted by Afshar et al. developed the Capsule Network (CapsNet) architecture for brain tumor classification using MRI images, which included the rough boundary of the tumor as an additional input, and achieved an accuracy of 90.89%, showing improved performance over conventional CapsNet and CNN in identifying meningioma, glioma, and pituitary tumors.[16]

The research study by Attique Khan et al. of deep learning-based brain tumor classification system using transfer learning techniques on VGG16 and VGG19 models, with ELM-based robust feature selection process and PLS-based feature fusion, which achieved an accuracy of 98.16% in the BRATS2015 dataset, 97.26% in the BRATS2017 dataset, and 93.40% in the BRATS2018 dataset, showed high efficiency in multimodal brain tumor classification.[17]

Research conducted by M. Badža et al. developed a new CNN architecture for brain tumor classification using T1-weighted contrast-enhanced MRI images, which achieved 96.56% accuracy on an enhanced dataset with 10-fold cross-validation, demonstrating high efficiency in the classification of meningioma, glioma, and pituitary tumors.[18]

Research conducted by Kumar et al. developed a Deep CNN-based Dolphin-SCA method for segmentation and classification of brain tumors using MRI images, which achieved 96.3% accuracy on BRATS and SimBRATS datasets, demonstrating high efficiency in identifying tumors with a good level of speed and accuracy. [19]

The research conducted by H. Sultan et al. developed a special CNN architecture for the classification of brain tumors based on MRI images, achieving 96.13% accuracy on the classification of meningioma, glioma, and pituitary tumors, as well as 98.7% accuracy on the classification of gliomas based on grade, demonstrating high ability in multi-classification of brain tumors.[20]

The research study by Kang et al. developed an MRI-based brain tumor classification method using ensemble deep features from 13 pre-trained CNN models and 9 types of machine learning classification, achieving an accuracy of 98.83% on datasets with two classes (normal and tumor) and 91.58% on datasets with four classes (normal, glioma, meningioma, and pituitary tumors), showing a significant improvement in classification performance through the ensemble learning approach.[21]

These studies aim to improve the accuracy and efficiency of segmentation and classification of brain tumors using methods based on deep learning, machine learning, transfer learning, and hybrid techniques on MRI images. In addition, this research also seeks to overcome the limitations of small datasets, provide real-time diagnosis, and support medical personnel in clinical decision-making through the development of artificial intelligence-based diagnostic tools.

Existing research shows significant differences when using CNN architecture and other transfer learning techniques to classify medical images, such as brain kanke detection. While the results are quite encouraging, there are some precautions to consider. First, many studies use unstable datasets, which means the results may not be common to more widespread populations or countries in the world. In addition, even if the experimental conditions are met with a high degree of accuracy, practical implementation in clinical settings is not widely carried out.

Additionally, while data augmentation techniques have been used in several studies to improve model performance, they can also lead to overfitting if not implemented correctly. High computing power is required to develop complex models that can be implemented in a variety of environments. A more comprehensive assessment using relevant metrics and a broader comparative analysis is still needed. Finally, the issue of data drift and real-time implementation capabilities also needs to be examined more closely to ensure that the developed model can be effectively applied in real-world clinical scenarios.

III. METHOD

This method is used to predict and classify Brain Cancer using the Convolutional Neural Network (CNN) architecture, namely Densenet201, VGG19 and ResNet50 using the Multi Cancer Brain Cancer dataset from Kaggle.com as shown in Figure 1

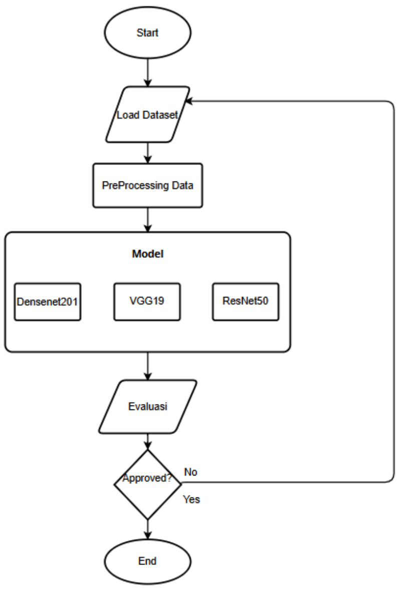

Figure 1. Flowchart Analysis Flow

Data Collection

The dataset used is Multi Cancer - Brain Cancer, which includes three different classes: three types of brain tumors: Meningioma, Glioma, and Pituitary tumor, which are processed for the classification of brain cancer types. The dataset can be viewed at the link https://figshare.com/articles/dataset/brain_tumor_dataset/1512427) This brain tumor dataset contains 3064 T1-weighted contrast-enhanced images from 233 patients with three types of brain tumors: meningioma (708 slices), glioma (1426 slices), and pituitary tumors (930 slices). An example of the Brain Cancer dataset image can be seen in figure 2. The Source code use can be accessed at the gituhub link: https://github.com/Rennuaqsha/Code_BrainCancer.git

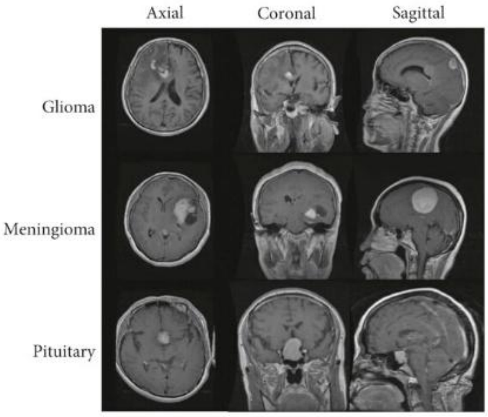

Figure 2. Three types of Brain Cancer

All of those in Figure 2 in our dataset (T1-weighted contrast-enhanced MRI) were acquired after Gd-DTPA injection at Nanfang Hospital, Guangzhou, China and General Hospital, Tianjin Medical University, China from 2005.9 to 2010.10. The image has a resolution in a 512 × 512 field with pixel dimensions of 0.49 × 0.49 mm^2. The thickness of the slice is 6 mm and the slice gap is 1 mm. The Gd dose is 0.1 mmol/kg at a rate of 2 ml/s.

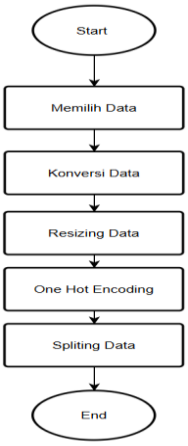

Figure 3. PreProcessing Flow

A. Data Preparation

This study aims to predict and classify Brain Cancer into three classes, namely Meningioma, Glioma, and Pituitary tumors using the Convolutional Neural Network (CNN) architecture based on DenseNet201. The dataset used was an image (T1-weighted contrast-enhanced MRI) taken from the Kaggle.com site, with a total of 3064 processed to train and test the model. The stages of data preprocessing can be seen in Figure 3.

The process in the image shows that the data processing in this study is carried out through several main stages, as depicted in the flowchart. The first stage is Selecting Data, where the multi-cancer dataset is downloaded from Kaggle using the kagglehub library and loaded using the load_files function. The dataset consists of several categories of cancer, and to ensure the efficiency and diversity of the data, as many as 800 images were randomly selected from the total dataset using the np.random.choice function. This aims to reduce computation time without compromising the representation of the dataset.

The next stage is Data Conversion, where the selected image file path is converted into a numerical array using the load_img and img_to_array functions. This function converts each image into a three-dimensional array that represents the pixel values in RGB format. This process is important to make the image in a format that can be processed by the deep learning model.

Next, Resizing Data is carried out for the standard image size to the dimension of 224x224 pixels. This size was chosen because it is an input resolution that is compatible with popular deep learning models such as DenseNet201 used in this study. Resizing is done directly during the conversion process using the target_size parameters on the load_img function.

The fourth stage is One-Hot Encoding, where the target label that was originally in the form of an integer is converted to a one-hot encoding format using the to_categorical function. In this format, each label is converted to a binary vector with a value of 1 on the index corresponding to its class, while the other indices are filled with a value of 0. This is done so that the model can process labels in the context of multi-class classification.

Table 1. Splitting Data

Train	Valid	Test
640	160	800

The last stage is the splitting of the data seen in Table 1, which divides the dataset into three subsets: training data, validation data, and flat test. This division is carried out using the train_test_split function with a ratio of 40:10:50 for training, validation and test data. This process ensures that the model is trained using training data, tested for performance on validation data, and evaluated ultimately using test data to avoid overfitting and measure the model's generalization objectively. This entire process aims to prepare the data for use in deep learning model training with an appropriate format and ensure optimal data distribution to support good model performance.

B. *Architectural Model*

The data processing process and model training begins with data preparation which includes steps such as Initializing, Transfer Learning, Complication Model and Model Training. The process can be seen in figure 4.

The process of processing data in code in the form of a flow like in an image, which involves several main steps. The process begins by selecting data from the available datasets, where the image datasets are downloaded from external sources using the Kaggle API. Once the data has been successfully downloaded, the next step is data conversion.

In this stage, the image data is converted into a numerical array using functions such as img_to_array to allow for further manipulation and analysis as shown in Figure 4. Then, the data is resizing to a uniform size, namely 224x224 pixels, to match the input requirements of pretrained models such as VGG19, ResNet50, or DenseNet201

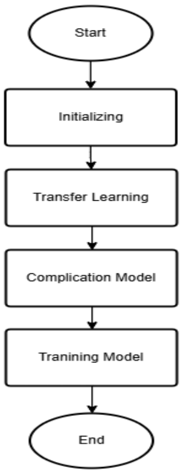

Figure 4. Model Architecture Drawings

Furthermore, a one-hot encoding process is applied to the label to convert the target value into a binary vector representation, which is required for classification tasks with more than one class. After that, the dataset is divided into three main parts through the data splitting process, namely training data, validation, and test data, with the aim of ensuring that the model can learn effectively from the training data while maintaining the generalization ability in the test data. This process ends by ensuring that the data is ready to be used for training transfer learning-based models.

The process begins with the Initializing step. At this stage, the system or model is prepared, including the initial configuration, parameter settings, and data setup to be used in the next process. Once the initialization process is complete, the next step is Transfer Learning, where a pre-trained model on a large dataset is used as a basis. Transfer learning allows for efficiency in model training, especially if the specific dataset used is relatively small.

The next stage is the Complication Model, which refers to the process of compiling a model by adding layers, parameters, or special architectures to customize the model in order to solve the specific problem at hand. This step is important to ensure that the model not only relies on the built-in features of transfer learning, but is also capable of handling the complexity of new data.

Once the model is compiled, the final step is Model Training. At this stage, the model is trained using a dataset that has been prepared. The training process involves optimizing model parameters to minimize errors and improve prediction performance. Once the training is complete, the process is terminated, and the model is ready for use. The models used are DenseNet 201, ResNet-50 and VGG19.

1. DenseNet (Dense Convolutional Network)

   DenseNet is a convolutional neural network architecture that introduces a unique approach in layer linking. In DenseNet, each layer is directly connected to all subsequent layers in the same network block.[22] The main advantage of DenseNet is dense connectivity, where each layer is directly connected to all subsequent layers, allowing for feature reuse, better gradient distribution, and a reduction in the number of parameters without compromising accuracy. This approach makes DenseNet very suitable for use on a limited number of datasets, as it can reduce the risk of overfitting. DenseNet is often used in medical and industrial applications, including in medical image analysis such as detecting cancer or other abnormalities, as well as in transfer learning for a variety of machine learning tasks. The uniqueness of densenet lies in the direct connection between layers within the network, where each layer will receive input from all previous layers. This provides the advantage of increased information flow and gradients within the network, so that densenet is able to achieve good performance with a relatively large number of parameters. The first block has six layers, the second block has twelve layers, the third block has forty-eight layers, and the fourth block has thirty-two layers. Each block is followed by a transition layer that includes 1x1 convolution for dimension reduction and 2x2 average pooling to reduce feature resolution. Finally, the Global Average Pooling (GAP) layer is used to reduce the feature to a single vector, which is passed to the Dense layer with a softmax activation function for multi-class classification. For this study, the initial layer of DenseNet-201 was frozen to maintain the pre-trained weights on the ImageNet dataset, and a special classification head was added that included the GAP, Dropout (with a level of 0.2), and the Dense layer.

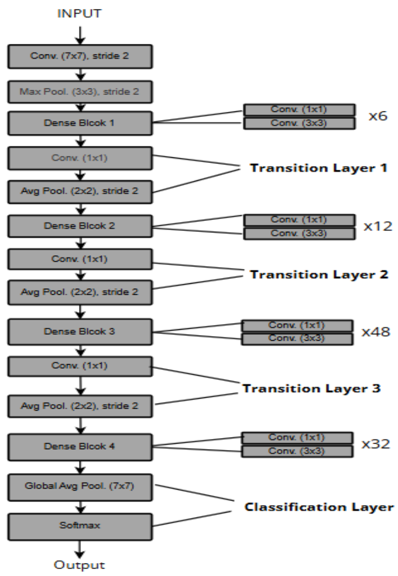

Figure 5. DenseNet20 Model Architecture

2. ResNet-50 (Residual Network-50)

ResNet-50 is a deep learning architecture designed to solve the problem of vanishing gradients in very deep neural networks, by introducing the concept of residual learning through shortcut connections or skip connections. This architecture consists of five main residual blocks, where the first block has 3 layers, the second block has 4 layers, the third block has 6 layers, and the fourth block has 3 layers, each equipped with a convolutional layer (1x1 and 3x3 kernels), batch normalization, and ReLU activation. After each block, the initial input is summed up with the transform output to allow for better gradient propagation and reduce the risk of overfitting. The model uses Global Average Pooling (GAP) to reduce features to a single vector, followed by Dropouts with a level of 0.2, and a Dense layer with a softmax activation function for multi-class classification. In this study, the initial layer of ResNet-50 was frozen to maintain the pre-trained weights on the ImageNet dataset, while a custom classification head was added to support more specific learning tasks. The main advantage of the

ResNet-50 lies in its parameter efficiency and ability to handle a limited number of datasets, making it ideal for applications such as medical image classification, object detection, and transfer learning in various domains.

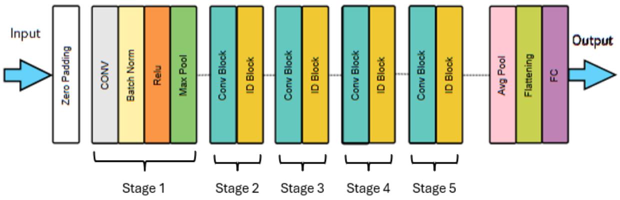

Figure 6. Model Architecture. ResNet50

3. VGG19

VGG19 is a Convolutional Neural Network (CNN) architecture designed by the Visual Geometry Group (VGG) of the University of Oxford for image recognition tasks, with a total of 19 layers consisting of 16 convolution layers, 3 fully connected layers, and one layer

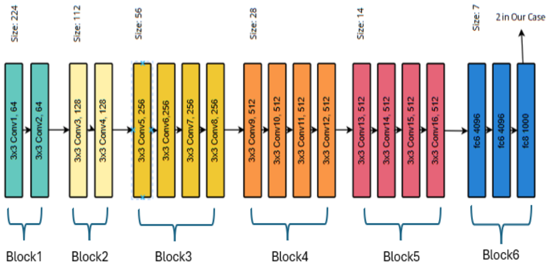

Figure 7. VGG19 model architecture

of softmax for classification. This model uses a 3x3 filter with same padding and 2x2 pooling to maintain simplicity while improving feature extraction. The VGG19 accepts 224x224x3 input images and has the advantages of a simple, modular, and powerful design, although it requires large resources with about 143 million parameters, which makes it computationally heavy and prone to overfitting if training data is limited.

C. Evaluation Matrix

In the process of developing machine learning models, model performance evaluation is a crucial step to ensure the effectiveness and accuracy of predictions. Evaluation metrics are used to measure how well a model is at predicting new data and provide an in-depth picture of the model's performance in various aspects. This sub-chapter will discuss the various metrics used to evaluate the model's results, such as accuracy, precision, recall, F1-score, and confusion matrix, which help in understanding the strengths and weaknesses of the model that has been built.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \times 100 \quad (1)$$

$$Sensitivity = \frac{TP}{TP+FN} \times 100 \quad (2)$$

$$Specificity = \frac{TN}{TN+FP} \times 100 \quad (3)$$

$$Accuracy = \frac{TP}{TP+FP} \times 100 \quad (4)$$

$$F1 - SCcore = 2 \times \frac{Sensitivity \times Precision}{Sensitivity + Precision} \times 100\% \quad (5)$$

1. Accuracy

Accuracy is a measure of how much of the predicted data is correct overall. This metric is often used as an early indicator of model performance because it includes correct predictions for both positive and negative classes.

2. Sensitiviy

Sensitivity, or Recall, measures a model's ability to identify positive data from all data that is actually positive. This metric is important in cases such as disease detection or fraud, where failure to detect positive cases has serious consequences.

3. Specifity

Specifity measures the model's ability to find negative data from all the actual negative data. This metric shows how well the model finds negative samples.

4. Precision

Precision shows how accurate the positive predictions generated by the model are, i.e. the proportion of correct positive predictions of all positive predictions. High precision indicates that the model rarely generates false positive predictions.

5. F1-Score

F1-Score is an evaluation metric that combines Precision and Recall in a single value by calculating the harmonized average of both. This metric balances the model's ability to detect positive data (Recall) and generate accurate positive predictions (Precision).

IV. IMPLEMENTASI

The implementation chapter in this study describes the process of implementing coding in the google colab environment, with library and runtime settings. The model architecture in this code adopts a transfer learning approach using the DenseNet201, ResNet-50 and VGG19 models. DenseNet201 is one of the convolutional neural network (CNN) architectures that connects each layer to all previous layers through a direct connection. This approach allows for a better flow of information within the network, reduces the risk of losing important information, and improves parameter efficiency. The model accepts a 224x224 pixel image input with three color channels (RGB), which is rescaled so that the value is in the range of 0 to 1.

In this implementation, the initial layer of DenseNet201 that has been trained on the ImageNet dataset is frozen to retain the basic features that have been learned earlier. Then, an additional layer is added on top of it to adapt the model to the classification task on the multi-cancer dataset. The additional layer includes GlobalAveragePooling2D, which flattens the output of DenseNet201 to reduce the feature dimensionality. After that, a Dropout layer with a dropout rate of 20% is added to reduce overfitting by disabling a certain number of neurons during training. The last layer is the Dense layer with a softmax activation function that generates probabilities for each class in the dataset.

This model is compiled using Adam optimizer with a small learning rate (0.0001) to ensure stable learning. The loss function used is binary_crossentropy, because the dataset involves classification with two classes. Data augmentation is applied to the training data using an ImageDataGenerator, including shear, zoom, and horizontal flip, to improve the generalization of the model. The model is trained by leveraging ModelCheckpoint callbacks to store the best model during training based on accuracy on validation data. After the training is complete, the model is evaluated to calculate the accuracy and loss on the test data. The process also includes prediction visualization and confusion matrix analysis to assess the model's performance in image classification. This approach demonstrates how the efficient

DenseNet201 can be adapted to solve specific classification tasks by leveraging transfer learning and regulatory techniques such as Dropout to optimize performance and reduce overfitting.

Furthermore, it utilizes the ResNet50 pre-trained model. ResNet50 is one of the convolutional neural network (CNN) architectures that is well-known for its ability to handle gradient vanishing problems through the use of residual blocks. The model is designed to process a 224x224 pixel image input with three color channels (RGB). In this implementation, the initial layer of pre-trained ResNet50 on the ImageNet dataset is frozen up to the 15th layer to retain the basic features that have been learned.

After that, an additional layer is added on top of the base model to adapt the model to the classification task on the multi-cancer dataset. The additional layers include GlobalAveragePooling2D to even out the output of the convolution layer, a Dropout layer with a dropout rate of 20% to reduce the risk of overfitting, and a Dense layer as the final layer with a softmax activation function, which generates probabilities for each class in the dataset.

This model is compiled using Adam optimizer with a learning rate of 0.0001 to ensure stable learning. The loss function used is binary_crossentropy, which is suitable for binary classification tasks. To improve model performance, data augmentation is performed using ImageDataGenerator, such as shear, zoom, and horizontal flip. The model is trained with a ModelCheckpoint callback to store the best weights based on validation accuracy during the training process.

After the training is complete, the model is evaluated using test data to calculate accuracy and loss. Further analysis is carried out with confusion matrix visualization and classification reports to evaluate the prediction performance on the test data. This approach leverages the power of ResNet50 as an efficient backbone for extracting visual features, while additional layers ensure the model can adapt to specific datasets. Regulatory techniques such as Dropout and data augmentation are applied to improve model generalization and prevent overfitting.

Table 22. result run with 4 time DenseNet201

Model	Optimizer	LR	Batch Size	Epoch
DenseNet201 (95)	Adam	0.0001	32	5
DenseNet201 (99)	Adam	0.0001	32	5
ResNet50(37)	Adam	0.0001	32	5
VGG19(91)	Adam	0.0001	32	5

DenseNet201 is one of the convolutional models used in this experiment. The model is designed for efficiency by utilizing the direct connection between the layers, thus allowing for a better flow of information. DenseNet201 is trained using the ImageNet dataset and can be adapted for image classification tasks through transfer learning. The model takes input in the form of a 224x224 pixel image with three color channels (RGB) and processes it through layers of convolution to extract important visual features. After that, the 4D output of the model is flattened using GlobalAveragePooling2D before being passed to the Dense layer to generate class predictions. Dropout layers with a dropout rate of 50% are applied to reduce the risk of overfitting during training. This model is compiled using Adam optimizer with a learning rate of 0.0001 and binary crossentropy as a loss function.

In the table above, there are several DenseNet201 configurations with the same optimizer, learning rate, batch size, and epoch settings. The first configuration uses DenseNet201 with a batch size of 32, a learning rate of 0.0001, and trains the model for 5 epochs. The second configuration with a batch size of 32 epochs The third configuration with the number of epochs is 5 to compare the results with the previous configuration. The fourth configuration maintains the same settings, namely Adam optimizer, learning rate 0.0001, batch size 32, and 5 epochs, for consistency of analysis.

V. RESULT AND DISCUSSION

In this section, the results of experiments conducted using three deep learning model architectures, namely DenseNet201, ResNet50, and VGG19, will be discussed in depth. Each model was tested for the task of classifying three classes of brain cancer images, namely brain_glioma, brain_menin, and brain_tumor, with predefined parameters, such as learning rate, Adam optimizer, batch size, and number of epochs. Discussions included analysis of accuracy and loss graphs, model performance based on confusion matrix, and the strengths and weaknesses of each architecture in handling these datasets. These results provide a comprehensive overview of the effectiveness and efficiency of the three models in multi-class image classification tasks.

A. Densenet201

In this experiment, the DenseNet201 model with a learning rate of 0.0001 showed excellent performance in the classification task of three classes: brain_glioma, brain_menin, and brain_tumor. The model was trained using Adam optimizer, with a batch size of 32 and a number of epochs of 5. The training results showed near-perfect training accuracy and very high validation accuracy, with consistency between the accuracy of the training and validation data, indicating good convergence without overfitting.

The accuracy graph in the image below shows significant progress during training. In the first epoch, the validation accuracy was around 90%, but as the number of epochs increased, the accuracy increased closer to 98% by the end of the training. The accuracy of the training data even reaches 99.50%, reflecting that the model successfully learns the pattern very well from the training data. There was no significant difference between training accuracy and validation, suggesting that the model had a strong generalization.

The loss graph in Figure 8 shows a consistent downward trend in losses for both training and validation data. At the initial epoch, the validation loss value is slightly higher than the training data loss, but decreases rapidly as the epoch increases. At the end of the training, the loss of validation data was almost close to the loss of training data. This indicates that the model learns patterns efficiently without overfitting

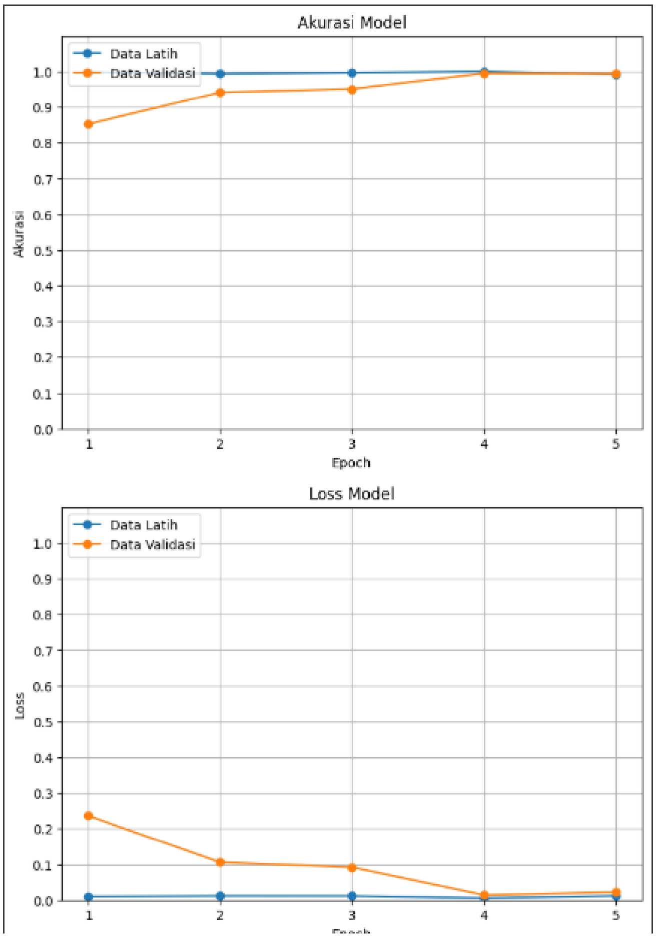

Figure 8. Accuracy and Loss of DenseNet201 Model

The confusion matrix in Figure 9 shows the prediction performance of the model on the test data. From the total test data, the model successfully predicted 280 brain_glioma data, 263 brain_menin data, and 253 brain_tumor data correctly. There was only 1 brain_glioma prediction error, 2 brain_menin prediction errors, and 1 brain_tumor prediction error. This error is very small compared to the amount of data correctly predicted, indicating that the model has very high sensitivity and precision for all three classes.

Based on these results, the DenseNet201 model with a learning rate configuration of 0.0001, Adam optimizer, and 5 epochs provides optimal results in this three-class image classification task. The combination of high accuracy, low loss, and minimal prediction error shows that this model is well suited for use in multi-class image classification on this dataset. Strong model generalizations on validation data also ensure that the model can work well on data that has never been seen before.

B. ResNet50

In the next experiment, there is a ResNet50 model with a learning rate of 0.0001 showing quite interesting training results in three class classification tasks: brain_glioma, brain_menin, and brain_tumor. The model was trained using Adam optimizer, with a batch size of 32 and a number of epochs of 5.

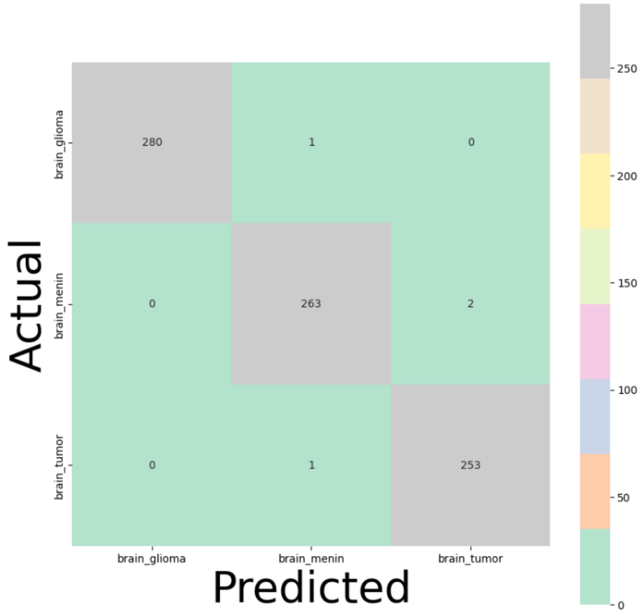

Figure 9. DenseNet201 Confusion Matrix Image

The accuracy graph shown in Figure 10 shows that the model managed to achieve the accuracy of the training data showing an accuracy of 37.25% at the end of the training. However, the accuracy of the validation data remained stagnant at around 35%-40%, which indicates that the model is overfitting. The model manages to learn patterns from the training data very well but is unable to capture the relevant patterns in the validation data, so its performance decreases on data that has never been seen before.

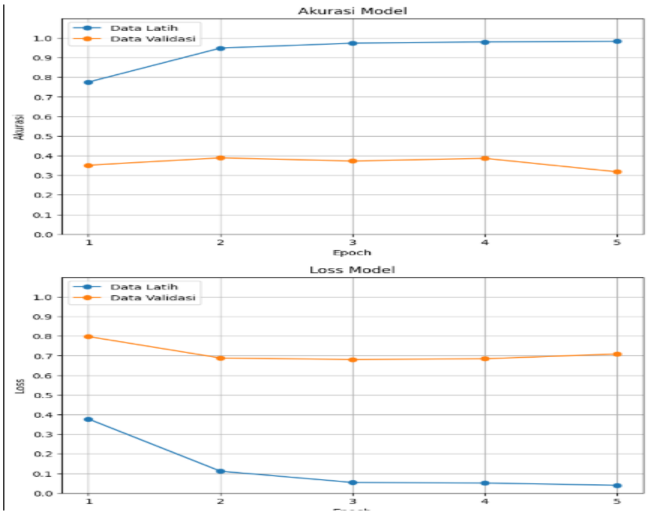

Figure 10. ResNet50 Accuracy and Loss

The loss graph supports this finding, where the loss value of the training data decreases significantly to close to zero, while the loss of validation data remains high and does not experience a significant decrease. This indicates that the model is too focused on the training data without paying attention to generalizations to validation data.

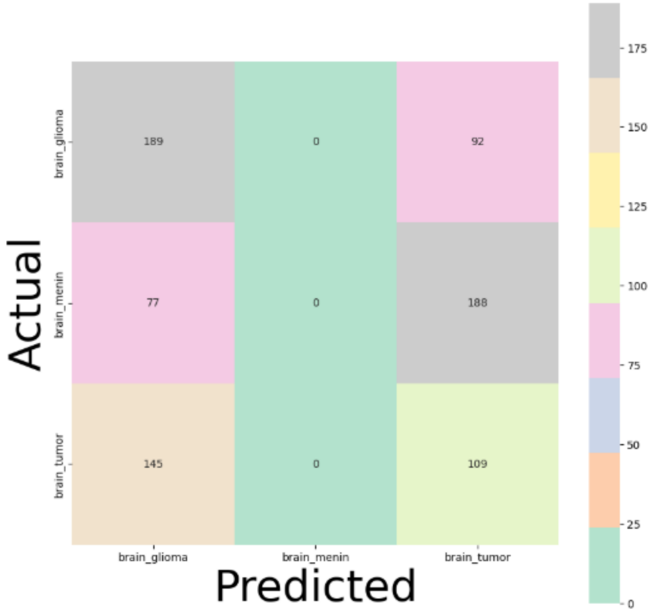

Figure 11. Confusion Matrix ResNet50

The confusion matrix in Figure 11 shows the performance of the model on the test data. The model correctly predicted 189 brain_glioma data, 188 brain_menin data, and 109 brain_tumor data. However, there are significant prediction errors, such as 92 errors in brain_glioma that are predicted as brain_tumor, and 145 errors in brain_tumor that are also predicted as brain_glioma. This error indicates that although the model can learn patterns from the training data very well, its generalization capabilities are still very limited.

Based on these results, it can be concluded that the ResNet50 model with a learning rate configuration of 0.0001, Adam optimizer, and 5 epochs is not able to provide optimal performance for this dataset due to overfitting that occurs. To improve generalization, it is necessary to implement strategies such as stronger regularization, additional data augmentation, or even the exploration of other models that are more suitable for these datasets. This is important to ensure that the model is not only good at training data but can also provide accurate predictions on validation and test data.

C. Visual Geometry Group (VGG19)

In this experiment, the VGG19 model was used for the classification of three class images with parameters such as a learning rate of 0.0001, an Adam optimizer, a batch size of 32, and a number of epochs of 5. Based on the training results, the model showed a significant increase in accuracy in both training data and validation data.

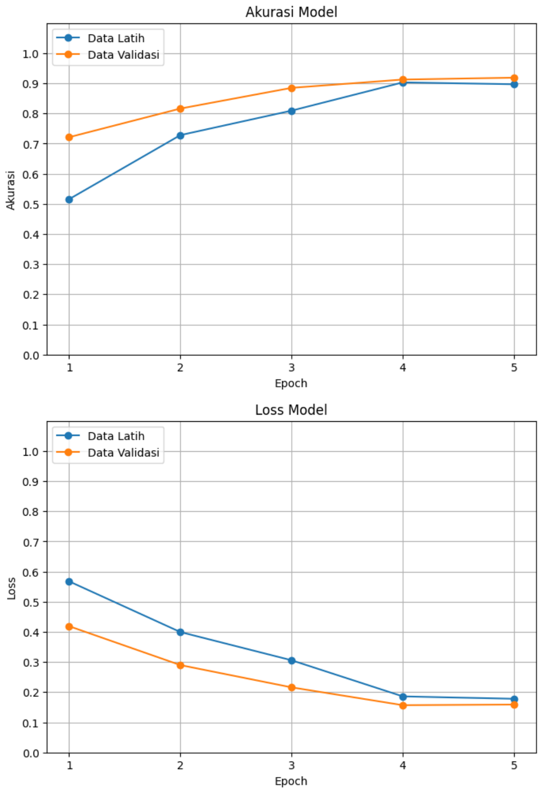

Figure 12. VGG19 Accuracy and Loss

The accuracy graph in Figure 12 shows that in the first epoch, the validation accuracy reaches about 0.8 and continues to increase until it approaches the training accuracy at the end of the training, which is about 0.9. The loss graph shows a consistent decline in training and validation data, reflecting the model's ability to learn effectively without any signs of overfitting.

The confusion matrix as shown in figure 8 illustrates the model's performance in predicting three classes, namely brain_glioma, brain_menin, and brain_tumor. The model correctly predicted 264, 222, and 244 samples in each class. However, there was a prediction error where some samples from class brain_menin were predicted to be brain_tumor as many as 42 cases, and samples from class brain_glioma were predicted to be wrong as many as 17 cases in total.

The results of Figure 13 below show that the VGG19 model has a good performance in studying the patterns in the dataset. While there are still prediction errors, the high accuracy achieved, as well as consistent accuracy and loss graphs, demonstrate the potential of VGG19 for more complex image classification tasks. With more optimal parameter settings or additional training data, the performance of this model can be further improved.

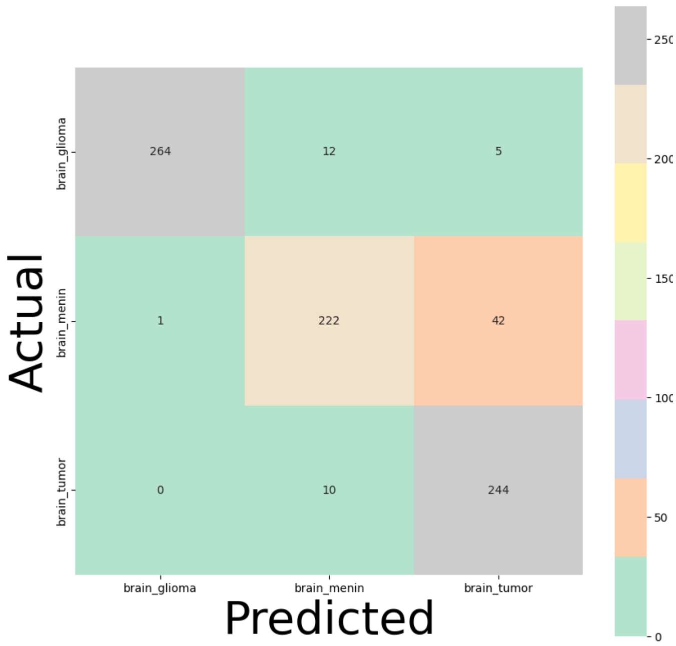

Figure 13. Confusion Matrix VGG19

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The conclusion of this study shows that the DenseNet201, ResNet50, and VGG19 models have different characteristics and performance in the task of classifying three classes of brain cancer images: brain_glioma, brain_menin, and brain_tumor. DenseNet201 shows the best performance with a training accuracy of 99.50% and a validation accuracy of nearly 98%, as well as a consistent decrease in losses without any signs of overfitting. The model also has strong generalizations, with a minimal number of prediction errors in the confusion matrix. In contrast, ResNet50 is overfitting, where validation accuracy stagnates at 35%-40%, even though the training accuracy reaches 37.25%. A significant prediction error indicates that this model is less able to capture relevant patterns in the test data. The VGG19 model provides quite good results with validation accuracy close to training accuracy, which is around 90%, as well as a stable loss trend without overfitting. However, there is a higher prediction error compared to DenseNet201, especially in the brain_menin class which is often mispredicted as brain_tumor. Overall, DenseNet201 is the most superior model for this dataset, while ResNet50 requires training strategy refinement, and VGG19 shows good potential with improvement opportunities through parameter optimization or additional data augmentation.

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